THE DIMENSION OF HARMONIC MEASURE ON GALTON–WATSON TREES INCREASES AT SMALL BIAS

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Abstract. Let T be a Galton–Watson tree without leaves whose offspring number N is not constant and has finite mean m, and let d_λ be the Hausdorff dimension of the harmonic measure of the λ-biased random walk on T, $0 \le \lambda < m$. Lyons, Pemantle and Peres asked whether d_λ increases with λ. We prove that $\lambda \mapsto d_\lambda$ is continuously differentiable on $[0, k_0)$, where k_0 is the smallest value taken by N, and that

$$\frac{d}{d\lambda} d_\lambda \Big|_{\lambda=0} = -\big(1 - \mathbb{E}[1/N]\big)\,\mathrm{Cov}\big(\log N, 1/N\big) > 0.$$

Hence d_λ is strictly increasing on an interval $[0, \varepsilon]$, with $\varepsilon > 0$ depending on the offspring law. No moment assumption beyond the finite mean is needed. The proof combines the formula of Lin and Rousselin for d_λ with a contraction estimate, uniform over trees, for the recursion satisfied by the escape probabilities and by their derivatives in λ.

1. Introduction

Let $(p_k)_{k\ge1}$ be a probability distribution on the positive integers with $p_k < 1$ for every k and finite mean $m = \sum_k k p_k$, so that $m > 1$, and let N denote a random variable with this distribution. Let T be a Galton–Watson tree with offspring distribution (p_k), rooted at e. Every vertex of T has at least one child. We write $\nu(x)$ for the number of children of a vertex x and

$$k_0 = \min\{k \ge 1 :\ p_k > 0\}$$

for the smallest possible number of children.

For $\lambda \ge 0$ the λ-biased random walk on T starts at the root, moves from the root to a uniformly chosen child, and moves from a vertex $x \ne e$ to its parent with probability $\lambda/(\lambda + \nu(x))$ and to each of its children with probability $1/(\lambda + \nu(x))$. Lyons [2] proved that the walk is transient exactly when $\lambda < m$. In that case the walk converges to a random ray of T; the law of this ray, given T, is the λ-harmonic measure on the boundary ∂T of T. The boundary carries the metric $d(\xi, \eta) = e^{-|\xi \wedge \eta|}$, where $|\xi \wedge \eta|$ is the number of edges common to the rays ξ and η. Lyons, Pemantle and Peres [3, 4] proved that the Hausdorff dimension of the λ-harmonic measure is almost surely equal to a constant d_λ, and that $d_\lambda < \log m$, the dimension of ∂T. For $\lambda = 0$ the walk never backtracks, the harmonic measure is the visibility measure of [3], and

$$d_0 = \mathbb{E}[\log N].$$

Lyons, Pemantle and Peres [5] asked whether d_λ is increasing in λ on $[0, m)$, and whether it is strictly increasing; this is Question 17.29 of [6]. Lin [1] and Rousselin [7] found, independently, an explicit stationary measure for the environment seen along the harmonic ray, and with it a formula for d_λ, recalled in Section 3. Lin deduced that $d_\lambda > d_0$ for every $\lambda \in (0, m)$, that $d_\lambda \to d_0$ as $\lambda \to 0$ and that $d_\lambda \to \log m$ as $\lambda \to m$. The comparison of d_λ and $d_{\lambda'}$ for $0 < \lambda < \lambda' < m$ is left open in [1], and the numerical computations of [7] support monotonicity.

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Theorem 1. *The function $\lambda \mapsto d_\lambda$ is continuously differentiable on $[0, k_0)$, with the one-sided derivative at 0, and*

$$(1)\quad d_0' = \big(1 - \mathbb{E}[1/N]\big)\Big(\mathbb{E}[1/N]\,\mathbb{E}[\log N] - \mathbb{E}\big[N^{-1}\log N\big]\Big) = -\big(1 - \mathbb{E}[1/N]\big)\,\mathrm{Cov}\big(\log N, 1/N\big).$$

This number is strictly positive.

Corollary 2. *There is $\varepsilon > 0$, depending on the offspring distribution, such that $d_\lambda < d_{\lambda'}$ whenever $0 \le \lambda < \lambda' \le \varepsilon$.*

Corollary 2 answers the question of [5], including strictness, on a neighbourhood of $\lambda = 0$. Since $1/N$ is bounded and $\mathbb{E}[\log N] \le m < \infty$, the right-hand side of (1) is finite under the standing assumptions; no second moment is used anywhere.

Example 3. If N takes the values 1 and 2 with probability $1/2$ each, then $\mathbb{E}[1/N] = 3/4$, $\mathbb{E}[\log N] = (\log 2)/2$ and $\mathbb{E}[N^{-1}\log N] = (\log 2)/4$, so $d_0' = (\log 2)/32$. If N takes the values 2 and 4 with probability $1/2$ each, then $d_0' = 5(\log 2)/128$. For the geometric distribution $p_k = 2^{-k}$ one has $\mathbb{E}[1/N] = \log 2$ and $d_0' = 0.05496$ to five decimal places.

Remark 4. In the formula of Lin and Rousselin, d_λ is the mean of $\log(\lambda + C_\lambda(T))$ under a probability measure on trees with an explicit density W_λ with respect to the Galton–Watson measure; here $C_\lambda(T)$ is the effective conductance of T. Both the function and the density depend on λ. At $\lambda = 0$ the derivative of $\log(\lambda + C_\lambda(T))$ is $\big(1 - \sum_{i \le \nu(e)} 1/\nu(e_i)\big)/\nu(e)$, where $e_1, \dots, e_{\nu(e)}$ are the children of the root. Its mean is zero, and it is negative with positive probability: this happens when the root has more than k_0 children and each of them has k_0 children. The slope (1) therefore comes entirely from the change of the density, which to first order favours trees whose root has many children; see (8).

The escape probability of the walk from a tree is the fixed point of a recursion over the subtrees of the children of the root. Section 2 shows that, written for the logarithm of the escape probability and restricted to trees in which every vertex has at least k children, this recursion is a contraction in the supremum norm with constant λ/k, uniformly in the tree, and that the same holds for the recursion obtained by differentiating in λ. This gives continuous differentiability of the escape probability of every such tree on $[0, k)$, with a bound on the derivative that does not depend on the tree, and allows differentiation of the formula for d_λ under the expectation. Section 3 contains the proof of Theorem 1, and Section 4 reports a numerical evaluation of d_λ for four offspring distributions.

2. Escape probabilities

For an integer $k \ge 1$ let $\mathcal{T}_k$ be the set of rooted trees in which every vertex has finitely many and at least k children. For $T \in \mathcal{T}_1$ let T_* be the tree obtained by joining a new vertex e_* to the root e of T, so that e_* is the parent of e. For $\lambda \ge 0$ consider the walk on T_* that starts at e and moves from every vertex $x \ne e_*$ to its parent with probability $\lambda/(\lambda + \nu(x))$ and to each child with probability $1/(\lambda + \nu(x))$. For $j \ge 0$ let

$$b_j(T, \lambda) = \mathbb{P}\big(\text{the walk reaches generation } j \text{ of } T \text{ before } e_*\big),$$
$$\beta_\lambda(T) = \mathbb{P}\big(\text{the walk never visits } e_*\big),$$

so that $b_0 = 1$. If the root of T has n children, we write $T_1, \dots, T_n$ for the subtrees of T rooted at these children; they belong to $\mathcal{T}_k$ when T does.

Lemma 5. *Let $T \in \mathcal{T}_1$, $\lambda \ge 0$ and $j \ge 1$, and let $S = \sum_{i=1}^{n} b_{j-1}(T_i, \lambda)$. Then*

$$b_j(T, \lambda) = \frac{S}{\lambda + S}, \tag{2}$$

and $b_j(T, \lambda)$ decreases to $\beta_\lambda(T)$ as $j \to \infty$.

Proof. The first j generations of T form a finite set. For $\lambda > 0$ the walk leaves this set almost surely, because from each of its vertices the walk reaches e_* in at most j steps with probability bounded below; for $\lambda = 0$ the walk moves away from the root at every step. Hence, almost surely, the walk reaches e_* or generation j. Condition on the first step, which leads to e_* with probability $\lambda/(\lambda + n)$ and to the ith child with probability $1/(\lambda + n)$. From the ith child the walk reaches generation j of T before returning to e with probability $b_{j-1}(T_i, \lambda)$, and otherwise it returns to e and starts afresh. Therefore $b = b_j(T, \lambda)$ satisfies

$$b = \frac{1}{\lambda + n} \sum_{i=1}^{n} \Big(b_{j-1}(T_i, \lambda) + \big(1 - b_{j-1}(T_i, \lambda)\big) b \Big),$$

that is $b(\lambda + S) = S$. Here $\lambda + S > 0$: this is clear if $\lambda > 0$, and if $\lambda = 0$ then every b_{j-1} equals 1 and $S = n$. So (2) holds. The events in the definition of b_j decrease with j, and by the first sentence of the proof their intersection is, up to a null event, the event that the walk never visits e_*. □

Lemma 6. *Let $k \ge 1$, $T \in \mathcal{T}_k$ and $0 \le \lambda \le k$. Then $b_j(T, \lambda) \ge 1 - \lambda/k$ for every $j \ge 0$, hence $\beta_\lambda(T) \ge 1 - \lambda/k$. If $\lambda < k$, then*

$$\beta_\lambda(T) = \frac{C_\lambda(T)}{\lambda + C_\lambda(T)}, \qquad C_\lambda(T) = \sum_{i=1}^{n} \beta_\lambda(T_i). \tag{3}$$

Proof. The bound holds for $j = 0$. If it holds for $j - 1$ and all trees of $\mathcal{T}_k$, then in (2) we have $S \ge n(1 - \lambda/k) \ge k - \lambda$, and since $s \mapsto s/(\lambda + s)$ is nondecreasing, $b_j(T, \lambda) \ge (k - \lambda)/k$. Letting $j \to \infty$ in (2), which involves finitely many subtrees and a denominator bounded below by $k - \lambda > 0$, gives (3). □

For $\lambda > 0$ the quantity $C_\lambda(T)$ is the effective conductance between the root of T and infinity when an edge between generations j and $j + 1$ has conductance λ^{-j}.

Proposition 7. *Let $k \ge 1$ and $0 < r < k$, and put $c = 1 - r/k$ and $M = k/(k - r)^2$. For every $T \in \mathcal{T}_k$ the function $\lambda \mapsto \beta_\lambda(T)$ is continuously differentiable on $[0, r]$, and for all $\lambda \in [0, r]$,*

$$c \le \beta_\lambda(T) \le 1, \qquad \big|\partial_\lambda \beta_\lambda(T)\big| \le M, \qquad \partial_\lambda \beta_\lambda(T)\big|_{\lambda = 0} = -\frac{1}{\nu(e)}.$$

Proof. Throughout, T ranges over $\mathcal{T}_k$ and λ over $[0, r]$, and $\|\cdot\|$ denotes the supremum over all such pairs. Put $\rho = r/k < 1$ and $x_j = \log b_j$. By Lemma 6, $\log c \le x_j \le 0$.

The recursion in logarithmic coordinates. For $n \ge k$ and $x \in [\log c, 0]^n$ let $S(x) = \sum_i e^{x_i}$ and $H_\lambda(x) = \log S(x) - \log(\lambda + S(x))$, so that by (2)

$$x_j(T, \lambda) = H_\lambda\big(x_{j-1}(T_1, \lambda), \dots, x_{j-1}(T_n, \lambda)\big).$$

We have

$$\frac{\partial H_\lambda}{\partial x_i} = q\, w_i, \qquad q = \frac{\lambda}{\lambda + S}, \quad w_i = \frac{e^{x_i}}{S}, \qquad \frac{\partial H_\lambda}{\partial \lambda} = -\frac{1}{\lambda + S}. \tag{4}$$

The w_i are nonnegative and sum to 1. Since $S \ge nc \ge k - r$, we have $q \le r/(r + k - r) = \rho$ and $1/(\lambda + S) \le 1/(k - r)$. The cube $[\log c, 0]^n$ is convex, so H_λ is ρ-Lipschitz on it for

the maximum norm. Consequently $\delta_j = \|x_j - x_{j-1}\|$ satisfies $\delta_j \le \rho\,\delta_{j-1}$ for $j \ge 2$, and $\delta_1 = \sup \log((\lambda+n)/n) \le \log(1+\rho) \le \rho$. Thus $\delta_j \le \rho^j$.

The derivatives. For fixed T and j, the function $\lambda \mapsto b_j(T,\lambda)$ depends on the first j generations of T only and, by (2) and induction, is a rational function of λ that is positive on $[0,r]$. Let $y_j = \partial_\lambda x_j$, so that $y_0 = 0$. By the chain rule and (4),

$$y_j(T,\lambda) = -\frac{1}{\lambda + S_j} + q_j \sum_{i=1}^{n} w_{j,i}\, y_{j-1}(T_i,\lambda), \tag{5}$$

where S_j, q_j and $w_{j,i}$ are computed from the vector $(x_{j-1}(T_i,\lambda))_{i\le n}$. Hence $\|y_j\| \le 1/(k-r) + \rho\|y_{j-1}\|$, and by induction

$$\|y_j\| \le \frac{1}{(k-r)(1-\rho)} = M \qquad \text{for all } j.$$

For $j \ge 2$ subtract (5) for $j-1$ from (5) for j:

$$y_j - y_{j-1} = \Big(\frac{1}{\lambda+S_{j-1}} - \frac{1}{\lambda+S_j}\Big) + q_j \sum_i w_{j,i}\big(y_{j-1}-y_{j-2}\big)(T_i) + \sum_i \big(q_j w_{j,i} - q_{j-1} w_{j-1,i}\big) y_{j-2}(T_i).$$

Since $|e^u - e^v| \le |u-v|$ for $u, v \le 0$, we have $|b_{j-1}(T_i) - b_{j-2}(T_i)| \le \delta_{j-1}$ for every i and $|S_j - S_{j-1}| \le n\delta_{j-1}$. All of S_j, S_{j-1} are at least nc. The first bracket is therefore at most $n\delta_{j-1}/(nc)^2 \le \delta_{j-1}/(kc^2)$ in absolute value. The second term is at most $\rho\|y_{j-1} - y_{j-2}\|$. For the third, write $q\,w_i = \lambda b_i/(S(\lambda+S))$ with $b_i = e^{x_i}$, and let b_i', S' be the corresponding quantities one step earlier. Then

$$\begin{aligned}\sum_i \Big|\frac{b_i}{S(\lambda+S)} - \frac{b_i'}{S'(\lambda+S')}\Big| &\le \frac{\sum_i |b_i - b_i'|}{S(\lambda+S)} + S'\,\Big|\frac{1}{S(\lambda+S)} - \frac{1}{S'(\lambda+S')}\Big| \\ &= \frac{\sum_i |b_i - b_i'|}{S(\lambda+S)} + \frac{|S-S'|\,(\lambda+S+S')}{S(\lambda+S)(\lambda+S')} \le \frac{n\delta_{j-1}}{(nc)^2} + \frac{2n\delta_{j-1}}{(nc)^2},\end{aligned}$$

where the last step uses $\lambda + S + S' \le (\lambda+S) + (\lambda+S')$. Multiplying by $\lambda \le r$, the third term is at most $3r\delta_{j-1}M/(kc^2)$. Altogether, with $K = (1+3rM)/(kc^2)$ and $\eta_j = \|y_j - y_{j-1}\|$,

$$\eta_j \le \rho\,\eta_{j-1} + K\delta_{j-1} \le \rho\,\eta_{j-1} + K\rho^{j-1} \qquad (j \ge 2),$$

and $\eta_1 = \|y_1\| \le 1/(k-r)$. By induction $\eta_j \le \rho^{j-1}\big(\eta_1 + (j-1)K\big)$, so $\sum_j \eta_j < \infty$.

Conclusion. The functions y_j converge uniformly on $\mathcal{T}_k \times [0,r]$ to a function y with $\|y\| \le M$. For fixed T each $y_j(T,\cdot)$ is continuous on $[0,r]$, hence so is $y(T,\cdot)$. By Lemma 5, $x_j(T,\lambda) \to x(T,\lambda) = \log \beta_\lambda(T)$. Passing to the limit in $x_j(T,\lambda) = x_j(T,0) + \int_0^\lambda y_j(T,t)\,dt$, and noting that $x_j(T,0) = 0$, we get $x(T,\lambda) = \int_0^\lambda y(T,t)\,dt$. Thus $\lambda \mapsto \log \beta_\lambda(T)$ is continuously differentiable on $[0,r]$ with derivative $y(T,\cdot)$, and $|\partial_\lambda \beta_\lambda| = \beta_\lambda |y| \le M$. Finally, at $\lambda = 0$ we have $b_{j-1} \equiv 1$, $S_j = n$ and $q_j = 0$ in (5), so $y_j(T,0) = -1/n$ for every $j \ge 1$, and the same holds for $y(T,0) = \partial_\lambda \beta_\lambda(T)|_{\lambda=0}$. □

3. Proof of Theorem 1

Let T and T' be independent Galton–Watson trees with offspring distribution (p_k). Almost surely both belong to $\mathcal{T}_{k_0}$, as do all the subtrees rooted at their vertices. For $0 \le \lambda < k_0$ put

$$B_\lambda = \beta_\lambda(T'), \qquad C_\lambda = C_\lambda(T), \qquad W_\lambda = \frac{B_\lambda C_\lambda}{\lambda - 1 + B_\lambda + C_\lambda}, \qquad L_\lambda = \log(\lambda + C_\lambda).$$

For $0 < \lambda < m$ the same definitions make sense, with β_λ and C_λ related as in (3), and Lin [1, formulas (7) and (8) and the display that follows (9)] proved that

$$d_\lambda = \frac{\mathbb{E}[W_\lambda L_\lambda]}{\mathbb{E}[W_\lambda]} \qquad (0 < \lambda < m); \tag{6}$$

see also [7, Theorem 4.1]. The function $T \mapsto \mathbb{E}[W_\lambda \mid T]/\mathbb{E}[W_\lambda]$ is the density, with respect to the Galton–Watson measure, of a probability measure on trees that is stationary for the environment seen along the harmonic ray. At $\lambda = 0$ we have $\beta_0 \equiv 1$, hence $C_0 = \nu(e)$, $W_0 = 1$ and $L_0 = \log \nu(e)$, and the right-hand side of (6) is $\mathbb{E}[\log N] = d_0$. So (6) holds for $\lambda = 0$ as well.

Proof of Theorem 1. Fix $r \in (0, k_0)$ and let $c = 1 - r/k_0$ and $M = k_0/(k_0 - r)^2$. Write $n = \nu(e)$ for the number of children of the root of T and $D_\lambda = \lambda - 1 + B_\lambda + C_\lambda$. Primes denote derivatives in λ. By Proposition 7, applied to T' and to $T_1, \dots, T_n$, the functions B_λ and C_λ are continuously differentiable on $[0, r]$ and

$$c \le B_\lambda \le 1, \qquad cn \le C_\lambda \le n, \qquad |B'_\lambda| \le M, \qquad |C'_\lambda| \le nM.$$

By Lemma 6, $B_\lambda \ge 1 - \lambda/k_0$ and $C_\lambda \ge n(1 - \lambda/k_0)$, so

$$D_\lambda \ge \lambda\Big(1 - \frac{1}{k_0}\Big) + n\Big(1 - \frac{\lambda}{k_0}\Big) \ge cn, \qquad D_\lambda \le r + n.$$

Therefore W_λ and L_λ are continuously differentiable on $[0, r]$, with

$$W'_\lambda = \frac{B'_\lambda C_\lambda + B_\lambda C'_\lambda}{D_\lambda} - \frac{B_\lambda C_\lambda (1 + B'_\lambda + C'_\lambda)}{D_\lambda^2}, \qquad L'_\lambda = \frac{1 + C'_\lambda}{\lambda + C_\lambda},$$

and, using $n \ge 1$,

$$\frac{c^2}{1+r} \le W_\lambda \le \frac{1}{c}, \qquad |W'_\lambda| \le \frac{2M}{c} + \frac{1 + 2M}{c^2},$$

$$|L_\lambda| \le \log n + \kappa, \quad \kappa = \max\big(\log(1+r), \log(1/c)\big), \qquad |L'_\lambda| \le \frac{1+M}{c}.$$

Hence $|(W_\lambda L_\lambda)'|$ and $|W'_\lambda|$ are bounded on $[0, r]$ by a constant multiple of $1 + \log \nu(e)$, which is integrable because $\mathbb{E}[\log N] \le m$. By the mean value theorem and dominated convergence, $\lambda \mapsto \mathbb{E}[W_\lambda L_\lambda]$ and $\lambda \mapsto \mathbb{E}[W_\lambda]$ are continuously differentiable on $[0, r]$, with derivatives $\mathbb{E}[(W_\lambda L_\lambda)']$ and $\mathbb{E}[W'_\lambda]$. Since $\mathbb{E}[W_\lambda] \ge c^2/(1+r)$, the quotient (6) is continuously differentiable on $[0, r]$. As $r < k_0$ was arbitrary, this proves the first assertion, and

$$d'_\lambda = \frac{\mathbb{E}[W'_\lambda L_\lambda] + \mathbb{E}[W_\lambda L'_\lambda] - d_\lambda \, \mathbb{E}[W'_\lambda]}{\mathbb{E}[W_\lambda]} \qquad (0 \le \lambda < k_0). \tag{7}$$

It remains to evaluate (7) at $\lambda = 0$. Let $N = \nu(e)$ be the number of children of the root of T, let $N_1, \dots, N_N$ be the numbers of children of these children, let K be the number of children of the root of T', and put $A = \sum_{i \le N} 1/N_i$. By Proposition 7,

$$B_0 = 1, \quad B'_0 = -\frac{1}{K}, \qquad C_0 = N, \quad C'_0 = -A, \qquad D_0 = N, \quad D'_0 = 1 - \frac{1}{K} - A.$$

Substituting,

$$W'_0 = \frac{-N/K - A}{N} - \frac{N(1 - 1/K - A)}{N^2} = -\frac{1}{K} - \Big(1 - \frac{1}{K}\Big)\frac{1}{N}, \qquad L'_0 = \frac{1 - A}{N}; \tag{8}$$

the terms containing A cancel in W_0'. Let $a = \mathbb{E}[1/N]$ and $\ell = \mathbb{E}[\log N]$. The variable K is independent of N, and given N the N_i are independent copies of N, so $\mathbb{E}[A \mid N] = Na$. Hence

$$\mathbb{E}[L_0'] = a - a = 0, \qquad \mathbb{E}[W_0'] = -2a + a^2, \qquad \mathbb{E}[W_0' \log N] = -a\ell - (1-a)\mathbb{E}\big[N^{-1}\log N\big].$$

With $W_0 = 1$, $L_0 = \log N$ and $d_0 = \ell$, formula (7) gives

$$d_0' = -a\ell - (1-a)\mathbb{E}\big[N^{-1}\log N\big] + (2a - a^2)\ell = (1-a)\Big(a\ell - \mathbb{E}\big[N^{-1}\log N\big]\Big),$$

which is (1).

For the sign, let N and K be independent with law (p_k). Then

$$-2\,\mathrm{Cov}\big(\log N, 1/N\big) = \mathbb{E}\Big[\big(\log N - \log K\big)\Big(\frac{1}{K} - \frac{1}{N}\Big)\Big].$$

The integrand is nonnegative, since the logarithm is increasing and $x \mapsto 1/x$ is decreasing, and it is strictly positive on the event $\{N \neq K\}$, which has positive probability because N is not constant. Finally $a < 1$ because $\mathbb{P}(N \geq 2) > 0$. Hence $d_0' > 0$. □

Proof of Corollary 2. By Theorem 1 the derivative of d_λ is continuous on $[0, k_0)$ and positive at 0, so it is positive on some interval $[0, \varepsilon]$. □

Remark 8. Let Q_λ be the probability measure with density $W_\lambda/\mathbb{E}[W_\lambda]$ with respect to the law of (T, T'). Formula (7) reads

$$d_\lambda' = \mathbb{E}_{Q_\lambda}\Big[\frac{1 + C_\lambda'}{\lambda + C_\lambda}\Big] + \mathrm{Cov}_{Q_\lambda}\Big(\log(\lambda + C_\lambda), \frac{W_\lambda'}{W_\lambda}\Big) \qquad (0 \leq \lambda < k_0).$$

The derivatives B_λ' and C_λ' are nonpositive, since the escape probabilities decrease in λ, but this does not determine the sign of either term. At $\lambda = 0$ the first term vanishes and the second equals (1).

4. Numerical evaluation

Formula (6) can be evaluated by simulation of the recursion (3). A pool of 2×10^5 pairs, representing samples of $(\beta_\lambda(T), \partial_\lambda \beta_\lambda(T))$, is initialised at $(1, 0)$ and updated 300 times: every new pair is computed from (3) and its derivative in λ, with an independent offspring number and children drawn independently and uniformly from the current pool. The expectations in (6) and (7) are then replaced by averages over the pool. Each run was repeated with six seeds, and the same seeds were used for every λ.

The bias runs over the grid $\lambda/m \in \{0,\ 0.02,\ 0.05,\ 0.1,\ 0.2, \ldots, 0.9\}$, for the four offspring distributions of Tables 1 and 2.

offspring law	m	d_0	d_0'	pool average	quotient
$p_1 = p_2 = \frac{1}{2}$	1.500	0.34657	0.02166	0.02206 ± 0.00041	0.02228 ± 0.00041
$p_2 = p_4 = \frac{1}{2}$	3.000	1.03972	0.02708	0.02720 ± 0.00019	0.02718 ± 0.00019
$p_1 = p_2 = p_3 = \frac{1}{3}$	2.000	0.59725	0.04954	0.04925 ± 0.00020	0.04954 ± 0.00019
$p_k = 2^{-k}$	2.000	0.50783	0.05496	0.05572 ± 0.00062	0.05677 ± 0.00061

Table 1. The slope at $\lambda = 0$. The fourth column is (1), the fifth is the pool average of the right-hand side of (7) at $\lambda = 0$, and the last is $(d_\lambda - d_0)/\lambda$ at $\lambda = 0.02\,m$; the last two are means over six seeds with their standard errors.

In Table 1 the pool averages agree with (1) to within one and a half standard errors, and the difference quotients, taken at the positive bias $0.02\,m$, are within 0.002 of it. In Table 2 the

λ/m	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$p_1 = p_2 = \frac{1}{2}$	0.3466	0.3501	0.3540	0.3587	0.3643	0.3709	0.3787	0.3872	0.3954	0.4021
$p_2 = p_4 = \frac{1}{2}$	1.0394	1.0475	1.0556	1.0636	1.0713	1.0785	1.0850	1.0905	1.0948	1.0977
$p_1 = p_2 = p_3 = \frac{1}{3}$	0.5978	0.6080	0.6190	0.6310	0.6436	0.6561	0.6676	0.6775	0.6854	0.6909
$p_k = 2^{-k}$	0.5080	0.5203	0.5356	0.5548	0.5786	0.6056	0.6324	0.6559	0.6747	0.6878

TABLE 2. Estimates of d_λ, means over six seeds. The standard errors, over the whole grid, are at most 0.00050.

estimates increase along the grid and stay below $\log m$, which equals 0.4055, 1.0986, 0.6931 and 0.6931 for the four distributions. The increase holds for every one of the six seeds and every pair of consecutive grid points. This agrees with the computations of [7] and with a positive answer to the question of [5] on the whole interval $[0, m)$, which Theorem 1 leaves open for λ away from 0.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [10] and Math Brain [8]. In October 2026 AI agents working from the project's list of open problems in probability obtained Theorem 1 and proofs of it, and a further agent checked the arguments. In November 2026 a Claude Fable agent (Anthropic) checked the proof again, wrote the simulation program and prepared the manuscript. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Programs and outputs:
`https://github.com/yspennstate/harmonic-dimension-small-bias`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [8].

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